

Rational Weak-Null Witnesses Generate Every Non-Almost-Schur Filter

An explicit basis for the negation half of the problem in arXiv:0903.0659

Automated proof attempt
agent_lane_03 model: GPT5.6

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Status: substantial partial result, likely valid, pending human verification. A canonical rational family is an explicit basis for all filters that are *not* almost Schur, fully answering the second half of the source request. The requested basis for the decreasing almost-Schur class itself remains open.

Source problem and definitions

All filters below are free filters on $\mathbb{N}$. A filter $\mathcal{G}$ *dominates* $\mathcal{F}$ when $\mathcal{G} \supset \mathcal{F}$. The source calls $\mathcal{F}$ *almost Schur* if every sequence $(x_n) \subset \ell_1$ that converges weakly to zero along $\mathcal{F}$ has norms not separated from zero. Thus $\mathcal{F}$ is not almost Schur exactly when some weakly $\mathcal{F}$ -null sequence satisfies $\inf_n \|x_n\|_1 > 0$.

The non-almost-Schur property is increasing under domination. A subclass is a *basis* for an increasing class when its upward closure is the whole class. Section 6 of the source asks for explicit bases for the decreasing almost-Schur class and, separately, its increasing negation class [1]. The appended page-15 crop gives the exact wording.

Proof intuition

Failure of the almost-Schur property comes with a witness sequence. The filter contains every index set on which finitely many weak tests of that sequence are small. Those weak-neighborhood traces therefore generate a smaller canonical bad filter.

There seems to be uncountable analytic data in the witness, but its vectors can be normalized and approximated term by term by finitely supported rational unit vectors. The error tends to zero in norm, hence along every free filter. This turns every failure into a completely coded rational failure without changing any of its weak-neighborhood traces up to enlargement. Taking all such rational trace filters gives exactly the requested upward basis.

The canonical filters

Let

$$S_{\mathbb{Q}} = \{u \in c_{00}(\mathbb{Q}) : \|u\|_1 = 1\}.$$

Call a sequence $u = (u_n)_{n \in \mathbb{N}} \subset S_{\mathbb{Q}}$ *admissible* if for every $k, N \in \mathbb{N}$, every $\phi_1, \dots, \phi_k \in \ell_{\infty}$, and every $\varepsilon > 0$, the set

$$V(u; \phi_1, \dots, \phi_k, \varepsilon, N) = \{n \geq N : \max_{1 \leq j \leq k} |\phi_j(u_n)| < \varepsilon\} \quad (1)$$

is nonempty. Equivalently, zero belongs to the weak closure of every tail of (u_n) .

For admissible u , finite intersections of sets in (1) contain another such set: take the union of the finite test families, the smaller tolerance, and the larger tail index. Hence these sets generate a proper filter, denoted $\mathcal{W}_u$. It is free

because every generator lies beyond an arbitrary prescribed N . By construction,

$$u_n \longrightarrow_{\mathcal{W}_u} 0 \quad \text{weakly}, \quad \|u_n\|_1 = 1.$$

Thus $\mathcal{W}_u$ is not almost Schur.

Admissible sequences exist. Indeed, enumerate the countable dense set $S_{\mathbb{Q}}$ so that every member occurs infinitely often. Given finitely many functionals, their common kernel in ℓ_1 is nonzero; approximate a unit vector in that kernel by a member of $S_{\mathbb{Q}}$. A sufficiently accurate approximation lies in the prescribed finite weak neighborhood and recurs in every tail.

Exact basis theorem

Theorem 1 (Rational witness basis). *The class*

$$\mathfrak{B}_- := \{\mathcal{W}_u : u \in S_{\mathbb{Q}} \text{ is admissible}\}$$

is a basis for the increasing class of all non-almost-Schur filters. Equivalently, for every free filter $\mathcal{F}$,

$$\mathcal{F} \text{ is almost Schur} \iff \mathcal{W}_u \not\subset \mathcal{F} \text{ for every admissible } u. \quad (2)$$

Proof. We already observed that every $\mathcal{W}_u \in \mathfrak{B}_-$ is not almost Schur. Since that property is increasing, every filter dominating a member of $\mathfrak{B}_-$ is also not almost Schur.

Conversely, suppose $\mathcal{F}$ is not almost Schur. Choose a weakly $\mathcal{F}$ -null sequence $(x_n) \subset \ell_1$ and $\delta > 0$ such that $\|x_n\|_1 \geq \delta$ for every n . Put

$$y_n = \frac{x_n}{\|x_n\|_1}.$$

The scalar multipliers are bounded by $1/\delta$, so (y_n) is still weakly $\mathcal{F}$ -null and every $\|y_n\|_1 = 1$.

The rational finite-support unit sphere is norm dense in the unit sphere of ℓ_1 . Choose $u_n \in S_{\mathbb{Q}}$ with $\|u_n - y_n\|_1 \leq 1/n$. The error sequence is norm null in the ordinary sense and therefore norm null along the free filter $\mathcal{F}$. It follows that (u_n) is weakly $\mathcal{F}$ -null.

Every set in (1) therefore belongs to $\mathcal{F}$: finite weak convergence gives the same trace without the tail restriction, and freeness adds $\{n : n \geq N\}$. In particular, the sequence u is admissible and the filter generated by all its traces satisfies $\mathcal{W}_u \subset \mathcal{F}$. Hence every non-almost-Schur filter dominates a member of $\mathfrak{B}_-$, proving both the basis assertion and (2). $\square$

The construction is fully combinatorial if each u_n is viewed as a finite rational signed measure on $\mathbb{N}$ and each $\phi \in \ell_{\infty}$ as a bounded scalar sequence. No arbitrary vectors remain in the basis generators.

Why the positive half does not follow by maximal extension

Criterion (2) reduces the missing half to finding a useful explicit cofinal family of filters that simultaneously avoid every $\mathcal{W}_u$. Individual blockers exist, but their complements need not retain the finite-intersection property. The most obvious Zorn argument also fails because the almost-Schur class is not closed under unions of chains.

Proposition 2 (Chain obstruction under CH). *Assume the continuum hypothesis. There is a strictly increasing chain $(\mathcal{F}_{\alpha})_{\alpha < \omega_1}$ of countably based Schur filters whose union is not almost Schur.*

Proof. Fix an admissible u . The trace filter $\mathcal{W}_u$ has a base of cardinality at most the continuum: the finite test lists range over ℓ_{∞} , and the other parameters may be restricted to rational tolerances and integer tails. It cannot have a countable base, because the source proves that every countably based filter is strongly diagonal and hence Schur, whereas $\mathcal{W}_u$ is not even almost Schur [1, Corollary 4.5 and Theorem 3.5]. Under CH its character is therefore exactly ω_1 .

Enumerate a base as $(B_\xi)_{\xi < \omega_1}$. Recursively let $\mathcal{F}_{\alpha+1}$ be generated by $\mathcal{F}_\alpha$, B_α , and one further set $D_\alpha \in \mathcal{W}_u \setminus \mathcal{F}_\alpha$; at a countable limit take the union. Such a D_α always exists, since the filter generated before stage ω_1 has a countable base and therefore cannot equal $\mathcal{W}_u$. Every $\mathcal{F}_\alpha$ is a proper subfilter of $\mathcal{W}_u$, has a countable base, and hence is Schur by the cited source results. The choices of D_α make the chain strict, while the enumeration of the B_ξ gives

$$\bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha = \mathcal{W}_u,$$

which is not almost Schur, witnessed by (u_n) . □

The proposition does not show that no basis for almost-Schur filters exists; the source request permits many meanings of “explicit.” It shows sharply why maximal almost-Schur filters cannot simply be obtained by the standard chain-union form of Zorn’s lemma. Attempts through Schur dominators, block-respecting filters, diagonal filters, and simultaneous blockers all lead to the same unresolved extension issue. Accordingly no answer to the positive half is claimed.

Verification and literature boundary

The proof uses only normalization, norm-density of $S_{\mathbb{Q}}$, and the source’s countable-base Schur theorem for the optional CH obstruction. Each inclusion direction in Theorem 1 was checked directly from the definition of filter convergence. There is no computational dependency.

The bounded novelty audit searched the run indexes, the exact problem phrase, the exact title and citations, and the phrases “almost Schur filter” and “basis of filters” through 11 August 2026. Later filter-convergence papers were found, but no answer to this basis request or this rational witness basis. This is a bounded audit, not an exhaustive priority claim.

References

- [1] A. Avilés López, B. Cascales Salinas, V. Kadets, and A. Leonov, *The Schur ℓ_1 theorem for filters*, Zh. Mat. Fiz. Anal. Geom. **3** (2007), 383–398; arXiv:0903.0659. Included as `source_paper.pdf`.

Definition 6.7. Let $\mathcal{P}$ be an increasing (decreasing) class of filters. A class of filters $\mathcal{P}_1 \subset \mathcal{P}$ is said to be a *basis* for $\mathcal{P}$ if $\mathcal{P}$ is the smallest increasing (decreasing) class, containing $\mathcal{P}_1$.

The problem which looks interesting is to construct explicitly a class of almost Schur filters, which forms a base for the class of all almost Schur filters. The same question makes sense for the negation of property to be almost Schur filter. Such a study was done in [6] for the class of those filters $\mathcal{F}$, that weak $\mathcal{F}$ -convergence of a sequence implies existence of a bounded subsequence.